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Broadcast encryption with improved resource utilization

Abstract

A broadcast network can use a pairing-based broadcast encryption scheme for N users with $O(N.\sup.1/3)$ -sized parameters. A pairing-based ciphertext-policy attribute-based encryption (CP-ABE) scheme for the class of degree 3 polynomials can be implemented with compact parameters. The public key, ciphertext and secret keys comprise $O(n)$ group elements, where n is input length for the function. The constructions achieve adaptive security against unbounded collusions, and rely on the (bilateral) k-Lin assumption in prime-order bilinear groups.

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Background/Summary

(1) This application is a national stage application, filed under 35 U.S.C. § 371, of International Patent Application No. PCT/US2022/034089, filed Jun. 17, 2022, which claims the benefit of U.S. Provisional Application No. 63/212,633, filed Jun. 19, 2021, the contents of both of which are hereby incorporated in their entireties.

FIELD OF THE INVENTION

(1) The disclosure relates to pairing-based broadcast encryption schemes with $O(N^{\sup.1/3})$ -sized parameters, which achieves adaptive security against unbounded collusions.

BACKGROUND OF THE INVENTION

(2) In ciphertext-policy attribute-based encryption (CP-ABE), ciphertexts ct are associated with a predicate f and a message m and keys sk with an attribute x , and decryption returns m when x satisfies f . Broadcast encryption is a special case of CP-ABE where the predicate is specified by a set $S.Math.[N]$, and decryption returns m when $x \in S$. In both cases, we require security against unbounded collusions, so that an adversary that sees a ciphertext along with secret keys for an arbitrary number of attributes $x_{sub.1}$, $x_{sub.2}$, . . . learns nothing about m as long as none of these attributes satisfies f .

(3) Broadcast encryption has been an active area of research since its introduction in the 1990s, where a major goal is to obtain schemes with short parameters, notably short ciphertexts ct and short public keys mpk . Prior work includes a pairing-based broadcast encryption scheme with constant-size ciphertext (ignoring the contribution from the set S) and secret keys; however, the scheme has large public keys mpk which is linear in the total number of users N , and moreover, decryption requires access to mpk . To address these shortcomings, the scheme could be modified to achieve $O(\sqrt{N})$ -sized public keys, at the cost of a $O(\sqrt{N})$ -sized ciphertext. Further work showed how to achieve $O(\sqrt{N})$ -sized parameters (i.e., $|mpk|+|ct|+|sk|=O(\sqrt{N})$) under the standard k -Lin assumption, improving upon the q -type assumption, while additionally strengthening the security guarantees from selective to adaptive security.

(4) Other prior work constructed a broadcast encryption scheme with $\text{poly}(\log N)$ -sized parameters from pairings and LWE. Nonetheless, the following basic problem remains open: Can we build a broadcast encryption scheme with $o(\sqrt{N})$ -sized parameters (that is, $|mpk|+|ct|+|sk|=o(\sqrt{N})$) from (just) pairings?

(5) Prior approaches for pairing-based broadcast encryption requires $|ct|.Math.max\{|sk|, |mpk|\}=\Omega(N)$, which in turn implies a $\Omega(\sqrt{N})$ bound on the parameter size. Moreover, this is essentially optimal for a large class of approaches for pairing-based broadcast encryption, indicating that breaking the $\sqrt{N}$ barrier would require substantially new ideas.

BRIEF SUMMARY OF THE INVENTION

(6) The invention relates to broadcast encryption as well as attribute-based encryption schemes.

(7) The invention includes a pairing-based broadcast encryption scheme with $O(N^{\sup.1/3})$ -sized parameters, breaking the long-standing $\sqrt{N}$ barrier. The broadcast encryption scheme achieves adaptive security against unbounded collusions, and relies on the bilateral k -Lin assumption in prime-order bilinear groups. In addition, the construction offers a range of trade-offs between ciphertext and key sizes. Prior to this work, it was not known how to achieve $o(\sqrt{N})$ -sized parameters with selective security even with q -type assumptions or generic bilinear groups.

(8) More generally, the invention includes a CP-ABE for degree 3 polynomials over $\{0, 1\}^{\sup.n}$ (and more generally,  $\sup.p$ where the public key, ciphertext and secret keys comprise of $O(n)$ group elements; this scheme also achieves adaptive security against unbounded collusions under the bilateral k -Lin assumption. The broadcast encryption scheme then follows as an immediate corollary, since we can encode set membership in $S.Math.[N]$ as a degree 3 polynomial over $\{0, 1\}^{\sup.O(N^{\sup.1/3}.sup.)}$. Prior to this work, CP-ABE schemes with $O(n)$ -sized parameters from pairings was only known for the class of degree 2 polynomials.

(9) The design of the inventive schemes departs quite significantly from existing pairing-based ABE schemes, in that we exploit the power of “quadratic reconstruction”. This idea was previously used to construct an information-theoretic, private-key analogue of broadcast construction—formally, conditional disclosure of secrets (CDS) for index—with $O(N^{\sup.1/3})$ -sized parameters. However, the scheme only works over fields of characteristic 2, which are incompatible with bilinear groups operations “in the exponent”. Instead, the invention provides new techniques for instantiating quadratic reconstruction with reference to functional encryption for degree 2 polynomials.

(10) In various embodiments, a computer-implemented method, computing device, and computer-readable storage media are disclosed. In one example embodiment for encrypting a message for transmitting over a broadcast network to multiple recipients as a ciphertext, the computer-implemented method, computing device, and computer-readable storage media can comprise: receiving an authorized recipient set S at a broadcast module, wherein the authorized recipient set identifies a set of recipients with permission to recover a broadcast message, wherein S is a subset of $\{1, 2, \dots, N\}$, wherein n is an integer such that $n^{\sup.3} \geq N$, and wherein N is the total number of possible recipients of the message; generating a public key mpk at an authority, the public key mpk comprising three vectors $[v]_{sub.1}$, $[w]_{sub.1}$ and $[w]_{sub.2}$ of n random group elements in a first source group, a random group element $[v]_{sub.0}$ in the first source group, and a random group element $[\alpha]_{sub.T}$ in a target group; receiving the public key mpk at the broadcast module from the authority; receiving or creating a message M for encryption; encoding the authorized recipient set S as a degree 3 polynomial, wherein the degree 3 polynomial has a coefficient vector f equal to a characteristic vector of the recipient set S , and wherein the vector f has a length $n^{\sup.3}$; cryptographically compressing the vector f using the vector $[w]_{sub.2}$ of n group elements from the public key mpk and by sampling a random vector $w_{sub.3}$ of n integers modulo p resulting in a digest of n group elements; generating a ciphertext of message M by: sampling a random scalar s ; multiplying the random scalar s with the digest, the vector $w_{sub.3}$ and the public key mpk ; concatenating the output of the multiplying to form the ciphertext; and transmitting the ciphertext to the authorized recipients identified in the set S .

(11) Further embodiments can include generating the ciphertext by parsing $mpk=([w]_{sub.1}, [w]_{sub.2}, [v]_{sub.0}, [v]_{sub.1}, [\alpha]_{sub.T})$.

[α].sub.T); sampling $s \leftarrow \text{custom character.sub.p, w.sub.3} \leftarrow \text{custom character.sub.p.sub.n}$; and outputting ciphertext $ct = [s].\text{sub.1}, [(I.\text{sub.n}.\text{Math.w.sub.2}.\text{Math.w.sub.3})f.\text{sup.T} + w.\text{sub.1}.\text{sup.T}]s.\text{sub.1}, [w.\text{sub.3}s].\text{sub.1}, [(v.\text{sub.0}w.\text{sub.3} + v)s].\text{sub.1}, [as].\text{sub.T}.\text{Math.M}$.

(12) Further embodiments can include instructions for decrypting a ciphertext by a recipient to recover a broadcast message by receiving a ciphertext and an authorized recipient set at a receiver module from a remote broadcast module over a broadcast network, wherein the ciphertext further comprises a digest, and the digest further comprises information about a polynomial; receiving a key from an remote authority module, the key comprising group elements; checking whether the recipient is in a set of authorized recipients; computing the degree 3 polynomial that has a coefficient vector f equal to a characteristic vector of a recipient set S ; wherein the degree 3 polynomial has a coefficient vector f equal to a characteristic vector of the recipient set S , and wherein the vector f has a length $n.\text{sup.3}$; recovering the broadcast message by evaluating the polynomial with the digest using the group elements in a pairing algorithm; storing the recovered broadcast message for further processing at the receiver module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are included to provide further understanding and are incorporated in and constitute a part of this specification, illustrate disclosed embodiments, and together with the description, serve to explain the principles of the disclosed embodiments.
- (2) In the drawings:
- (3) FIG. 1 is a block diagram illustrating a multimedia communication system suitable for use in various embodiments.
- (4) FIG. 2 illustrates an example computer-implemented method for the claimed broadcast encryption scheme.
- (5) FIG. 3 illustrates an example computer system architecture for implementing the claimed systems and methods.
- (6) FIG. 4 illustrates further details of an example computer system architecture for implementing the claimed systems and methods.

DETAILED DESCRIPTION

(7) Technical Overview

(8) An overview of our constructions is disclosed herein. We focus on our CP-ABE scheme for degree 3 polynomials over $\text{custom character.sub.p.sub.n} \times \text{custom character.sub.p.sub.n} \times \text{custom character.sub.p.sub.n}$ given by $(x.\text{sub.1}, x.\text{sub.2}, x.\text{sub.3}) \in \text{custom character}(x.\text{sub.1}.\text{Math.x.sub.2}.\text{Math.x.sub.3}).\text{Math.f.sub.T}$ where $f \in \text{custom character.sub.p.sub.n.sub.3}$ is the coefficient vector. Throughout, we use boldface lower case to denote row vectors. In our CP-ABE scheme, encryption takes as input $f \in \text{custom character.sub.p.sub.n.sub.3}$ and a message M and outputs a ciphertext ct ; key generation takes as input $x.\text{sub.1}, x.\text{sub.2}, x.\text{sub.3} \in \text{custom character.sub.p.sub.n}$ and outputs a key sk ; and decryption takes as input ct, sk along with $f, x.\text{sub.1}, x.\text{sub.2}, x.\text{sub.3}$ and outputs M whenever $(x.\text{sub.1}.\text{Math.x.sub.2}.\text{Math.x.sub.3})f.\text{sup.T} \neq 0$.

(9) We rely on an asymmetric bilinear group $(\text{custom character.sub.1}, \text{custom character.sub.2}, \text{custom character.sub.T}, e)$ of prime order p where $e: \text{custom character.sub.1} \times \text{custom character.sub.2} \rightarrow \text{custom character.sub.T}$. We use $[\cdot].\text{sub.1}, [\cdot].\text{sub.2}, [\cdot].\text{sub.T}$ to denote component-wise exponentiations in respective groups $\text{custom character.sub.1}, \text{custom character.sub.2}, \text{custom character.sub.T}$. The k -Lin assumption in $\text{custom character.sub.1}$ asserts that $([A].\text{sub.1}, [sA].\text{sub.1}) \approx_{\text{sub.c}} ([A].\text{sub.1}, [u].\text{sub.1})$ where $s \leftarrow \text{custom character.sub.p.sub.k}, A \leftarrow \text{custom character.sub.p.sub.k} \times (k+1), u \leftarrow \text{custom character.sub.p.sub.k} - 0$. The bilateral k -Lin assumption as used in this work asserts that $([A].\text{sub.1}, [A].\text{sub.2}, [sA].\text{sub.2}) \approx_{\text{sub.c}} ([A].\text{sub.1}, [A].\text{sub.2}, [u].\text{sub.2})$, and is a strengthening of the k -Lin assumption in $\text{custom character.sub.2}$. In symmetric bilinear groups, the bilateral k -Lin and the standard k -Lin assumption are equivalent. Note that 1-Lin=DDH/SXDH, and that bilateral 1-Lin is false, for the same reason DDH is false in symmetric bilinear groups. We will describe our construction based the k -Lin assumption and the bilateral k' -Lin assumption, and set $k=1, k'=2$ for optimal concrete efficiency.

(10) We make extensive use of tensor products. This enables a more compact description of our schemes, and avoids triple summations to compute a degree 3 polynomial. Moreover, we will be replacing scalars with vectors as our schemes get increasingly complex, upon which some scalar-vector products translate naturally to a tensor product of two vectors, whereas some other ones translate to a vector-matrix product.

(11) Roadmap

(12) Herein is described two candidate CP-ABE schemes for degree 3 polynomials. The first achieves

$$|mpk| = O(n.\text{sup.2}), |ct| = O(n), |sk| = O(n)$$

(13) In comparison, prior constructions based on degree 2 polynomials require either $|ct| = O(n.\text{sup.2})$ or $|sk| = O(n.\text{sup.2})$. The second is a variant of the first with $|mpk| = O(n)$ and thus achieves $O(n)$ -sized parameters.

(14) Below it is described how to modify the second candidate to obtain our final CP-ABE scheme, which achieves $O(n)$ -sized parameters as well as adaptive security under the bi- k -Lin assumption.

(15) CP-ABE for Degree 2 Polynomials

(16) Disclosed is a simplified variant of the CP-ABE scheme for the class of degree 2 polynomials over $\text{custom character.sub.p.sub.n} \times \text{custom character.sub.p.sub.n}$ given by

$$(x.\text{sub.1}, x.\text{sub.2}) \in \text{custom character}(x.\text{sub.1}.\text{Math.x.sub.2}).\text{Math.f.sub.T}$$

where $f \in \text{custom character.sub.p.sub.n.sub.2}$ is the coefficient vector and decryption is possible whenever $(x.\text{sub.1}.\text{Math.x.sub.2})f.\text{sup.T} \neq 0$:

$$\begin{aligned} mpk &= [\quad], [w_2]_1, [w_1]_1, & w_1 &\leftarrow \mathbb{Z}_p^n, w_2 \leftarrow \mathbb{Z}_p^n, & &\leftarrow \mathbb{Z}_p \\ (17) \quad ct &= [s]_1, [(I_n.\text{Math.w}_2)^T + w_1^T]_1, [\quad]_T.\text{Math.M}, & s &\leftarrow \mathbb{Z}_p \\ sk &= [r]_2, [x_1 r w_1^T]_2, [x_2 \quad - r w_2]_2, & r &\leftarrow \mathbb{Z}_p \end{aligned} \quad (1)$$

(18) Note that the scheme achieves

$$|mpk| = O(n), |ct| = O(n), |sk| = O(n)$$

(19) Decryption uses

$$\begin{aligned} (x_1.\text{Math.x}_2)f^T.\text{Math.} \quad s &= (x_1.\text{Math.} ((x_2 \quad + r w_2)^{\text{ct}})f^T + \\ (20) \quad x_1 r.\text{Math.} ((I_{n_1}.\text{Math.w}_2).\text{Math.f}^T + w_1^T)s - & \\ x_1 r w_1^T.\text{Math.} \quad s & \end{aligned} \quad (2)$$

(21) Following a dual system encryption methodology, security boils down to showing that M is hidden given a single ciphertext-key pair. In particular, it suffices to show that if $(x.\text{sub.1}.\text{Math.x.sub.2})f.\text{sup.T} = 0$, then α is hidden given

$$(22) \quad \begin{aligned} \hat{ct} &= [(I_n \cdot \text{Math. } w_2) \cdot \text{Math. } f^\top + w_1^\top]_1, \\ \hat{sk} &= [x_1 w_1^\top]_2, [x_2 - w_2]_2, \end{aligned} \quad (3)$$

where custom character are derived from ct,sk by setting $r=s=1$ and omitting $[\alpha s]_{\text{sub.T}}$. Hiding of α then follows from

$$(23) \quad (\hat{ct}, \hat{sk}) \equiv (\tilde{w}_1^\top, ((x_1 \cdot \text{Math. } \tilde{w}_2) f^\top + x_1 w_1^\top - (x_1 \cdot \text{Math. } x_2) f^\top \cdot \text{Math. } , \tilde{w}_2))$$

Our First Candidate CP-ABE

(24) A candidate CP-ABE is disclosed for degree 3 polynomials with parameter sizes

$$|mpk| = O(n \cdot \text{sup.2}), |ct| = O(n), |sk| = O(n)$$

(25) To arrive at this scheme, we first replace $x_{\text{sub.2}}$ and $w_{\text{sub.2}}$ in (1) with $x_{\text{sub.2}} \cdot \text{Math. } x_{\text{sub.3}}$ and $w_{\text{sub.2}} \cdot \text{Math. } w_{\text{sub.3}}$ respectively, where $w_{\text{sub.3}} \leftarrow$ custom character.sub.p.sup.n. The ciphertext size remains unchanged, but the secret key size increases to $O(n \cdot \text{sup.2})$ due to the term $(x_{\text{sub.2}} \cdot \text{Math. } x_{\text{sub.3}}) \alpha - r(w_{\text{sub.2}} \cdot \text{Math. } w_{\text{sub.3}})$

(26) To achieve $|sk| = O(n)$, we compute the above expression using

$$(27) \quad x_2 \cdot \text{Math. } x_3 - r w_2 \cdot \text{Math. } w_3 = x_2 \cdot \text{Math. } (x_3 + r_3 w_3) - (x_2 r_3 + r w_2) \cdot \text{Math. } w_3$$

(28) This yields the following scheme:

$$(29) \quad \begin{aligned} mpk &= []_T, [w_1]_1, [w_3]_1, [w_2 \cdot \text{Math. } w_3]_1, w_1, w_2, w_3 \leftarrow \mathbb{Z}_p^n, \leftarrow \mathbb{Z}_p \\ ct &= [s]_1, []_T \cdot \text{Math. } M, [((I_n \cdot \text{Math. } w_2 \cdot \text{Math. } w_3) \cdot \text{Math. } f^\top + w_1^\top) s]_1, [w_3 s]_1, s \leftarrow \mathbb{Z}_p, \\ sk &= [r_2]_2, [x_1 r_2 w_1^\top]_2, [x_2 r_3 + r_2 w_2]_2, [x_3 + r_3 w_3]_2, r_2, r_3 \leftarrow \mathbb{Z}_p \end{aligned} \quad (4)$$

(30) Here, we publish $[w_{\text{sub.2}} \cdot \text{Math. } w_{\text{sub.3}}]_{\text{sub.1}}$ in mpk so that we can compute $[(w_{\text{sub.2}} \cdot \text{Math. } w_{\text{sub.3}}) s]_{\text{sub.1}}$ in ct.

(31) Compressing Mpk

(32) To get to a CP-ABE scheme with $O(n)$ -sized parameters, we compress mpk in the previous scheme as follows: instead of having set-up pick $w_{\text{sub.3}}$, the encryptor will sample a random $w_{\text{sub.3}}$; this eliminates $[w_{\text{sub.2}} \cdot \text{Math. } w_{\text{sub.3}}]_{\text{sub.1}}$ in mpk and reduces mpk to $O(n)$ group elements.

Next, we explain how this modification impacts ct and sk in (4): Given $[w_{\text{sub.2}}]_{\text{sub.1}}$, $w_{\text{sub.3}}$, s , f , it is easy to compute

$[(I_{\text{sub.n}} \cdot \text{Math. } w_{\text{sub.2}} \cdot \text{Math. } w_{\text{sub.3}}) \cdot \text{Math. } f \cdot \text{sup.T}]_{\text{sub.1}}$ and thus $[(I_{\text{sub.n}} \cdot \text{Math. } w_{\text{sub.2}} \cdot \text{Math. } w_{\text{sub.3}}) \cdot \text{Math. } f \cdot \text{sup.T} + w_{\text{sub.1}} \cdot \text{sup.T}]_{\text{sub.1}}$ in ct. Now, key generation can no longer compute $[x_{\text{sub.3}} \alpha + r_{\text{sub.3}} w_{\text{sub.3}}]_{\text{sub.2}}$, which was used to compute $[(x_{\text{sub.3}} \alpha + r_{\text{sub.3}} w_{\text{sub.3}}) s]_{\text{sub.2}}$ during decryption. Instead, we will compute the latter using the equation

$$(33) \quad (x_3 + r_3 w_3) s = (r_3 + r_2 v_0) \cdot \text{Math. } w_3^\top s + (x_3 + r_2 v) \cdot \text{Math. } s - r_2 \cdot \text{Math. } (v_0 w_3 + v) s \text{ where } v_{\text{sub.0}}, v \text{ are chosen by the set-up algorithm.}$$

(34) Putting these modifications together, the next candidate is obtained.

(35) Our Second Candidate CP-ABE

(36) The candidate CP-ABE scheme with $O(n)$ -sized parameters is disclosed, where the terms not present in the previous scheme are shaded in large curly braces:

(37)

$$mpk = []_T [w_2]_1, [w_1]_1, \{[v]_1, [v_0]_1\} w_1, w_2, v \leftarrow \mathbb{Z}_p^n, v_0 \leftarrow \mathbb{Z}_p \quad ct = [s]_1, []_T \cdot \text{Math. } M, [((I_n \cdot \text{Math. } w_2 \cdot \text{Math. } w_3) f^\top + w_1^\top) s]_1, [w_3 s]_1, \{[]_2, []_2\}$$

(38) The decryption algorithm on input $ct = ([s]_{\text{sub.1}}, [\alpha s]_{\text{sub.T}} \cdot \text{Math. } M, [c_{\text{sub.1}} \cdot \text{sup.T}]_{\text{sub.1}}, [c_{\text{sub.2}}]_{\text{sub.1}}, [c_{\text{sub.3}}]_{\text{sub.1}})$ and $sk = ([r_{\text{sub.2}}]_{\text{sub.2}}, [d_{\text{sub.1}}]_{\text{sub.2}}, [d_{\text{sub.2}}]_{\text{sub.2}}, [d_{\text{sub.3}}]_{\text{sub.2}}, [d_{\text{sub.4}}]_{\text{sub.2}})$, computes

$[(x_{\text{sub.1}} \cdot \text{Math. } x_{\text{sub.2}} \cdot \text{Math. } x_{\text{sub.3}}) f \cdot \text{sup.T} \cdot \text{Math. } \alpha s]_{\text{sub.T}}$ using

$$(39) \quad (x_1 \cdot \text{Math. } x_2 \cdot \text{Math. } (d_3 c_2 + d_4 s - r_2 c_3)) f^\top - () f^\top + (x_1 \cdot \text{Math. } x_2 \cdot \text{Math. } r_3 w_3) s + (x_1 \cdot \text{Math. } r_2 w_2 \cdot \text{Math. } w_3) s$$

where

$$(40) \quad \begin{aligned} (i) &= (r_3 + r_2 v_0) (w_3 s) + (x_3 + r_2 v) s - r_2 (v_0 w_3 + v) s = (x_3 + r_3 w_3) s \\ (ii) &= (x_2 r_3 + r_2 w_2) \cdot \text{Math. } (I_n \cdot \text{Math. } w_3 s) = (x_2 \cdot \text{Math. } r_3 w_3) s + (r_2 w_2 \cdot \text{Math. } w_3) s \\ (iii) &= x_1 r_2 ((I_n \cdot \text{Math. } w_2 \cdot \text{Math. } w_3) f^\top + w_1^\top) s = (x_1 \cdot \text{Math. } r_2 w_2 \cdot \text{Math. } w_3) f^\top s + x_1 r_2 w_1^\top s \\ (iv) &= x_1 r_2 w_1^\top s \end{aligned}$$

Our Final CP-ABE

(41) We now describe how we arrive at our final CP-ABE for the class of degree 3 polynomials, which achieves adaptive security against unbounded collusions under the k -Lin assumption in custom character.sub.1, custom character.sub.2 and the bilateral k' -Lin assumption, where $k \geq 1$, $k' \geq 2$.

We sample $A \leftarrow$ custom character.sub.p.sup. $(k+1) \times k$, $B \leftarrow$ custom character.sub.p.sup. $k \times (k+1)$ and make the following substitutions to the scheme in (??) combined with (??):

(42)

$$s \mapsto A s^\top \in \mathbb{Z}_p^{1 \times (k+1)}, k \in \mathbb{Z}_p^{k+1}, r \mapsto r B \in \mathbb{Z}_p^{k+1}, w_2 \mapsto W_2 \in \mathbb{Z}_p^{(k+1) \times (k+1)k'n}, w_1^\top \mapsto W_1 \in \mathbb{Z}_p^{(k+1)n \times (k+1)}, v \mapsto V \in \mathbb{Z}_p^{(k+1) \times (k+1)n}, v_0 \mapsto V_0 \in \mathbb{Z}_p^{(k+1) \times (k+1)n}$$

(43) That is, we increase the width and heights of each of $w_{\text{sub.2}}$, $w_{\text{sub.1}} \cdot \text{sup.T}$, v , $v_{\text{sub.0}}$ by a multiplicative factor of $k+1$. A complete description of the scheme is provided herein.

(44) Herein, a broader class of degree 3 polynomials over custom character.sub.p.sup. $n \cdot \text{sup.1} \times$ custom character.sub.p.sup. $n \cdot \text{sup.2} \times$

custom character.sub.p.sup. $n \cdot \text{sup.3}$ is described. By varying $n_{\text{sub.1}}$, $n_{\text{sub.2}}$, $n_{\text{sub.3}}$, we obtain trade-offs between ciphertext and key sizes.

(45) Notations

(46) We denote by $s \leftarrow S$ the fact that s is picked uniformly at random from a finite set S . We use $\approx_{\text{sub.s}}$ to denote two distributions being statistically indistinguishable, and $\approx_{\text{sub.c}}$ to denote two distributions being computationally indistinguishable. We use lower case boldface to denote row vectors and upper case boldface to denote matrices. For any positive integer N , we use $[N]$ to denote $\{1, 2, \dots, N\}$.

(47) Tensor Product

(48) The tensor product (Kronecker product) for matrices $A = (a_{\text{sub.i,j}}) \in$ custom character, $B \in$ custom character.sub. $n \times p$ is defined as

$$(49) \quad A \cdot \text{Math. } B = [] \cdot \text{Math. } , \cdot \text{Math. } , \cdot \text{Math. }] \in \mathbb{Z}^{n \times mp}.$$

(50) The mixed-product property for tensor product says that

$$(A \cdot \text{Math. } B)(C \cdot \text{Math. } D) = (AC) \cdot \text{Math. } (BD)$$

(51) A useful corollary of the mixed-product property says that for any pair of row vectors $u, v \in \mathbb{F}_p^{1 \times n}$

$$(52) \quad u \cdot \text{Math. } v = (u \cdot \text{Math. } 1)(I_n \cdot \text{Math. } v) = (1 \cdot \text{Math. } v)(u \cdot \text{Math. } I_n) \\ = u(I_n \cdot \text{Math. } v) = v(u \cdot \text{Math. } I_n)$$

(53) We adopt the convention that matrix multiplication takes precedence over tensor product, so that we can write $A \cdot \text{Math. } BC$ to mean $A \cdot \text{Math. } (BC)$.

(54) Prime-Order Bilinear Groups

(55) A generator $\mathbb{G}$ takes as input a security parameter $1.\text{sup.}\lambda$ and outputs a description $\mathbb{G} = (p, \mathbb{G}_1, \mathbb{G}_2, e)$, where p is a prime of $\Theta(\lambda)$ bits, $\mathbb{G}_1, \mathbb{G}_2$ and e are cyclic groups of order p , and $e: \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ is a non-degenerate bilinear map. We require that the group operations in $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ and the bilinear map e are computable in deterministic polynomial time in λ . Let $g_1 \in \mathbb{G}_1, g_2 \in \mathbb{G}_2$ and $g_T = e(g_1, g_2) \in \mathbb{G}_T$ be the respective generators. We employ the implicit representation of group elements: for a matrix M over $\mathbb{F}_p$, we define $[M]_1 = g_1.\text{sup.}M, [M]_2 = g_2.\text{sup.}M, [M]_T = g_T.\text{sup.}M$, where exponentiation is carried out component-wise. Also, given $[A]_1, [B]_2$, we let $e([A]_1, [B]_2) = [AB]_T$. We recall the matrix Diffie-Hellman (MDDH) assumption on $\mathbb{G}$:

(56) Assumption 1 (MDDH) Let $k, d \in \mathbb{N}$. We say that the MDDH assumption holds if for all PPT adversaries $\mathcal{A}$, the following advantage function is negligible in λ .

$$(57) \quad \text{Adv}_{\mathcal{A}}^{\text{MDDH}_{k,d}} := \Pr[(\mathbb{G}, [M]_1, [MS]_1) = 1] - \Pr[(\mathbb{G}, [M]_1, \{[U]_1\}) = 1] \cdot \text{Math.}$$

where $\mathbb{G} = (p, \mathbb{G}_1, \mathbb{G}_2, e) \leftarrow \mathbb{G}(1.\text{sup.}\lambda)$, $M \leftarrow \mathbb{F}_p^{k \times d}$, $S \leftarrow \mathbb{F}_p^{d \times d}$ and $U \leftarrow \mathbb{F}_p^{k \times d}$.

(58) The MDDH assumption on $\mathbb{G}$ can be defined in an analogous way. Escala et al. [?] showed that

$$k\text{-Lin.MDDH} \leq k \cdot \text{MDDH} \leq k \cdot k\text{-Lin.MDDH}$$

with a tight security reduction. (In the setting where $\mathbb{G}$ is a bilinear group, the MDDH assumption holds unconditionally.)

(59) The bilateral MDDH assumption is defined analogously with the advantage function:

$$(60) \quad \Pr[(\mathbb{G}, [M]_1, [MS]_1, [M]_2, [MS]_2) = 1] - \Pr[(\mathbb{G}, [M]_1, \{[U]_1\}, [M]_2, \{[U]_2\}) = 1] \cdot \text{Math.}$$

(61) Note that the bilateral MDDH and bilateral k -Lin assumptions are false for $k=1$. In the disclosed embodiments, we only require a weaker variant of the bilateral MDDH assumption, as defined with the advantage function:

$$(62) \quad \Pr[(\mathbb{G}, [M]_1, [M]_2, [MS]_2) = 1] - \Pr[(\mathbb{G}, [M]_1, [M]_2, \{[U]_2\}) = 1] \cdot \text{Math.}$$

Attribute-Based Encryption

(63) We define attribute-based encryption in the framework of key encapsulation. A attribute-based encryption scheme for a predicate $P(\cdot, \cdot)$ consists of four algorithms (Setup, Enc, KeyGen, Dec): Setup($1.\text{sup.}\lambda, \mathbb{G}$) outputs the public parameter mpk and the master key msk . Enc(mpk, x) outputs a ciphertext ct and a symmetric key $\text{kem} \in \{0, 1\}^{\text{sup.}\lambda}$. KeyGen(msk, y) outputs a secret key sk . Dec(sk, y, ct) outputs a symmetric key kem . It outputs a secret key sk . Dec(sk, y, ct) outputs a symmetric key kem . The decryption algorithm gets as input $\text{sk}, \text{ct}, x, y$ such that $P(x, y)=1$. It outputs a symmetric key kem .

(64) In the disclosed schemes, we can compute $\text{kem} \in \mathbb{F}_p$, which can then be hashed to $\{0, 1\}^{\text{sup.}\lambda}$.

(65) Correctness

(66) We require that for all $(x, y) \in \mathbb{F}_p \times \mathbb{F}_p$ such that $P(x, y)=1$,

$$\Pr[(\text{ct}, \text{kem}) \leftarrow \text{Enc}(\text{mpk}, x); \text{Dec}(\text{sk}, y, \text{ct}) = \text{kem}] = 1,$$

where the probability is taken over $(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1.\text{sup.}\lambda, \mathbb{G})$ and the coins of Enc.

Security Definition

(67) For a stateful adversary $\mathcal{A}$, we define the advantage function

$$(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1, \mathbb{G});$$

$$x \leftarrow \text{KeyGen}(\text{msk}, \cdot)(\text{mpk});$$

$$(68) \quad \text{Adv}_{\mathcal{A}}^{\text{ABE}} := \Pr[b = b' : b \leftarrow_R \{0, 1\}; \text{kem}_1 \leftarrow_R \{0, 1\}^{\text{sup.}\lambda}] - \frac{1}{2}$$

$$(\text{ct}, \text{kem}_0) \leftarrow \text{Enc}(\text{mpk}, x);$$

$$b' \leftarrow \text{KeyGen}(\text{msk}, \cdot)(\text{ct}, \text{kem}_0)$$

with the restriction that all queries y that $\mathcal{A}$ makes to $\text{KeyGen}(\text{msk}, \cdot)$ satisfies $P(x, y)=0$. An attribute-based encryption scheme is adaptively secure if for all PPT adversaries $\mathcal{A}$, the advantage $\text{Adv}_{\mathcal{A}}^{\text{ABE}}(\lambda)$ is a negligible function in λ .

CP-ABE for Degree 3 Polynomials

(69) Here,

$$\mathbb{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e) \text{ where } \mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G}_T = \mathbb{F}_p \text{ and } e(a, b) = ab \text{ for } a, b \in \mathbb{F}_p$$

and

$$P(f, (x_1, x_2, x_3)) = 1 \Leftrightarrow (x_1 + x_2 + x_3) \cdot f \neq 0$$

Broadcast Encryption

(70) Here,

$$\mathbb{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e) \text{ where } \mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G}_T = \mathbb{F}_p \text{ and } e(a, b) = ab \text{ for } a, b \in \mathbb{F}_p$$

where $\{0, 1\}^{\text{sup.}N}$ is considered to be the power set of $[N]$ (i.e., set of all subsets of $[N]$), and

$$P(S, y) = 1 \Leftrightarrow y \in S$$

CP-ABE for Degree 3 Polynomials

(71) Herein, we present an adaptively secure CP-ABE for degree 3 polynomials against unbounded collusions, under the k -Lin assumption in $\mathbb{G}$.

Our scheme achieves

$$|\text{mpk}| = (k(k+1) + k(k+1)(n.\text{sub.}1 + k'n.\text{sub.}2 + n.\text{sub.}3) + k') \cdot |\mathbb{G}_1| + |\mathbb{G}_T|$$

$$|\text{ct}| = (k+1 + (k+1)n.\text{sub.}1 + (k+1)k'n.\text{sub.}3 + (k+1)n.\text{sub.}3) \cdot |\mathbb{G}_1|$$

$$|\text{sk}| = (2k+1 + (k+1)k'n.\text{sub.}2 + (k+1)k' + (k+1)n.\text{sub.}3) \cdot |\mathbb{G}_2|$$

(72) Setting $k=1, k'=2$, we obtain

$$|\text{mpk}| = (2n.\text{sub.}1 + 4n.\text{sub.}2 + 2n.\text{sub.}3 + 4) \cdot |\mathbb{G}_1| + |\mathbb{G}_T|, |\text{ct}| = (2n.\text{sub.}1 + 6n.\text{sub.}3 + 2) \cdot |\mathbb{G}_1|$$

$$|\text{sk}| = (4n.\text{sub.}2 + 2n.\text{sub.}3 + 7) \cdot |\mathbb{G}_2|$$

The Scheme Setup($p, 1.\text{sup.}n.\text{sub.}1, 1.\text{sup.}n.\text{sub.}2, 1.\text{sup.}n.\text{sub.}3$): Run $\mathbb{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e)$

$\text{custom character.sub.T, e} \leftarrow \text{custom character(p). Sample}$
 $A \leftarrow \text{custom character.sub.p.sup.(k+1)} \times k, k \leftarrow \text{custom character.sub.p.sup.k+1, W.sub.2} \leftarrow \text{custom character.sub.p.sup.}$
 $(k+1) \times (k+1)k'.n.\text{sub.2, W.sub.1} \leftarrow \text{custom character.sub.p.sup.(k+1)n.sup.1.sup.} \times (k+1),$
 $V \leftarrow \text{custom character.sub.p.sup.(k+1)} \times (k+1)n.\text{sub.3, V.sub.0} \leftarrow \text{custom character.sub.p.sup.(k+1)} \times (k+1)k', B \leftarrow \text{custom character.sub.p.sup.k} \times (k+1)$
(73) For a matrix $M \in \text{custom character}$, we write $M := M(\text{custom character.Math.A}) \in \text{custom character}$. In particular, we have
 $k = kA, W.\text{sub.2} = W.\text{sub.2}(I.\text{sub.k}'n.\text{sub.2.Math.A}), W.\text{sub.1} = W.\text{sub.1}A, V = V(I.\text{sub.n.sub.3.Math.A}), V.\text{sub.0} = V.\text{sub.0}(I.\text{sub.k}'n.\text{sub.2.Math.A})$
Output
 $\text{mpk} = (\text{custom character, [A].sub.1, [k].sub.T, [W.sub.2].sub.1, [W.sub.1].sub.1, [V].sub.1, [V.sub.0].sub.1}), \text{msk} = (k, W.\text{sub.1}, W.\text{sub.2}, V.\text{sub.0}, B)$
 $\text{Enc}(\text{mpk}, f)$: Sample
 $s \leftarrow \text{custom character.sub.p.sup.k, W.sub.3} \leftarrow \text{custom character.sub.p.sup.k} \times n.\text{sub.3}$ and output
(74)

$$\text{ct} = \left(\left[\text{As}^\top \right]_{c_0^\top}, \left[(I_{n_1} \cdot \text{Math.} (\bar{W}_2(I_{n_2} \cdot \text{Math.} W_3 \cdot \text{Math.} s^\top)))f^\top + W_1 s^\top \right]_{c_1^\top}, \left[W_3 \cdot \text{Math.} \text{As}^\top \right]_{c_2}, \left[\bar{V}_0(W_3 \cdot \text{Math.} s^\top) + \bar{V}(I_{n_3} \cdot \text{Math.} s^\top) \right]_{c_3} \right), \text{kem} = [\bar{k}s^\top]_T$$

KeyGen(msk, x.sub.1, x.sub.2, x.sub.3): Sample
 $r.\text{sub.2} \leftarrow \text{custom character.sub.p.sup.k, r.sub.3} \leftarrow \text{custom character.sub.p.sup.(k+1)}k'$
and output
(75) $\text{sk} = ([r_2 B]_{d_0}, [(x_1 \cdot \text{Math.} r_2 B)W_1]_{d_1}, [(x_2 \cdot \text{Math.} r_3 + r_2 BW_2]_{d_2}, [(r_3 + r_2 BV_0]_{d_3}, [(x_3 \cdot \text{Math.} k + r_2 BV]_{d_4}) \text{Dec}(\text{sk}, (x.\text{sub.1}, x.\text{sub.2}, x.\text{sub.3}),$
 $\text{ct}, f)$: Output
(76) 0

$$[(x_1 \cdot \text{Math.} x_2 \cdot \text{Math.} (d_3 C_2 + d_4 (I_{n_3} \cdot \text{Math.} c_0^\top) - d_0 C_3))f^\top - (x_1 \cdot \text{Math.} (d_2 (I_{n_2} \cdot \text{Math.} C_2)))f^\top + (x_1 \cdot \text{Math.} d_0)c_1^\top - d_1 c_0^\top]_T$$

(i)
(ii)
(iii)
(iv)

where the terms in (i), (ii), (iii), (iv) are computed in $\text{custom character.sub.T}$ using the pairing.

Broadcast Encryption with Size $N.\text{sup.1/3}$

(77) We can encode broadcast encryption for N parties as CP-ABE for degree 3 polynomials whenever $n.\text{sub.1}n.\text{sub.2}n.\text{sub.3} \geq N$, by using the folklore encoding of set membership in $S.\text{Math.}[N]$ as a degree 3 polynomial over $\{0, 1\}.\text{sup.n.sup.1} \times \{0, 1\}.\text{sup.n.sup.2} \times \{0, 1\}.\text{sup.n.sup.3}$: given a set $S.\text{Math.}[N]$, let $f = (f.\text{sub.1}, \dots, f.\text{sub.N}) \in \{0, 1\}.\text{sup.N}$ denote the characteristic vector for the set S (that is, $f.\text{sub.i} = 1$ iff $i \in S$); given $y \in [N]$, we can pick $x.\text{sub.1} \in \{0, 1\}.\text{sup.n.sup.1}$, $x.\text{sub.2} \in \{0, 1\}.\text{sup.n.sup.2}$, $x.\text{sub.3} \in \{0, 1\}.\text{sup.n.sup.3}$ such that $x.\text{sub.1}.\text{Math.}x.\text{sub.2}.\text{Math.}x.\text{sub.3} \in \{0, 1\}.\text{sup.n.sup.1}.\text{sup.n.sup.2}.\text{sup.n.sup.3}$ is the characteristic vector of the set $\{y\}$; then, $(x.\text{sub.1}.\text{Math.}x.\text{sub.2}.\text{Math.}x.\text{sub.3})f.\text{sup.T} = 1$ iff $y \in S$.

(78) We can then set $n.\text{sub.1} = N.\text{sup.}\delta$, $n.\text{sub.2} = N.\text{sup.}1-2\delta$, $n.\text{sub.3} = N.\text{sup.}\delta$ for any $0 \leq \delta \leq 1/3$, which yields

$$|\text{mpk}| = O(N.\text{sup.}1-2\delta), |\text{ct}| = O(N.\text{sup.}\delta), |\text{sk}| = O(N.\text{sup.}1-2\delta)$$

(79) In particular, when $\delta = 1/3$, we achieve

$$|\text{mpk}| = O(N.\text{sup.}1/3), |\text{ct}| = O(N.\text{sup.}1/3), |\text{sk}| = O(N.\text{sup.}1/3)$$

File System Example Embodiment

(80) The inventive scheme can be configured for efficiency. We can instantiate our scheme with the popular BIS12-381 curve with $|\text{custom character.sub.1}|$ being 48 bytes and $|\text{custom character.sub.2}|$ being 96 bytes. File sharing in encrypted file systems is a non-limiting example of an application for the inventive broadcast encryption. The Windows EFS has a limit of 256 KB in the file header for the EFS meta-data, and supports a maximum of 800 individual users. Assuming 32-bit users IDs, we can support 1000 users with a file header (S, ct) of size $4 \times 1000 + 82 \times 48 = 7936$ bytes, where each user holds a secret key of size $67 \times 96 = 6432$ bytes. Results can be improved by setting $n.\text{sub.1} = 20$, $n.\text{sub.2} = 10$, $n.\text{sub.3} = 5$, which yields a header of size $4 \times 1000 + 72 \times 48 = 7456$ bytes and a secret key of size $57 \times 96 = 5482$ bytes. However, since $N = 1000$ is fairly small, the broadcast encryption scheme with $O(\sqrt{\text{square root over } (N)})$ parameters would also achieve similar performances: a file header of size $4 \times 1000 + 66 \times 48 = 7168$ and a secret key of size $68 \times 96 = 6528$ bytes.

(81) System Implementations

(82) The various embodiments may be implemented within a variety of communication systems, networks and/or mobile multi-media broadcast systems, an example of which is illustrated in FIG. 1. Specifically, FIG. 1 illustrates a communication system in which mobile receiver devices **102** may receive content from multimedia broadcast network **104**, unicast network **106**, or via the Internet **108**. A typical multimedia broadcast network **104** includes a plurality of broadcast transmitters **112** controlled by a mobile broadcast network control center/broadcast operation center (BOC) **114**. The multimedia broadcast network **104** broadcasts content from the broadcast transmitters **112** as mobile broadcast transmissions **113** for reception by the mobile receiver devices **102**. Within the BOC **114**, there may be one or more servers **110** for managing content broadcasts, and which provide a connection to the Internet **108**.

(83) In addition to the multimedia broadcast network **104**, mobile receiver devices **102** may communicate via a unicast network **106**, such as a cellular telephone network, WiFi network (not shown), WiMAX, etc. A typical cellular telephone network includes a plurality of cellular base stations **116** coupled to a network operations center **118**. The network operations center **118** operates to connect voice and data calls between mobile receiver devices **102** and other network destinations, such as via telephone land lines (e.g., a POTS network, not shown) and the Internet **108**.

(84) Communications between mobile receiver devices **102** and the unicast network **106** may be accomplished via two-way wireless communication links **115** such as LTE, 4G, 3G, CDMA, TDMA, and other cellular telephone communication technologies. Such two-way wireless communication links **115** may enable users to stream multimedia content to receiver devices (e.g., mobile devices).

(85) To facilitate Internet data communications (e.g., streaming video feeds), the unicast network **106** will typically include one or more servers **120** coupled to, or within, the network operations center **118** that provide a connection to the Internet **108**. Mobile receiver devices **102** may further connect to the Internet **108** via a wired connection when available, in which case the Internet **108** may serve as the unicast network. Mobile receiver devices **102** may also receive non-broadcast content over the Internet **108** using well known conventional web-based access protocols.

(86) Generally, the operations for receiving and rendering content by a receiver device (e.g., the mobile receiver devices **102** discussed above) may be divided into separate and independent groups or categories of operations, and each group or category of operations may be assigned to a layer (e.g., physical layer, data link layer, etc.). In each of these layers, various hardware and/or software components may implement functionality that is commensurate with responsibilities assigned to that layer. For example, media streams (e.g., broadcast, point-to-point, etc.) are typically received in the physical layer, which may include a radio receiver, buffers, and processing components that perform the operations of demodulating, recognizing symbols within the radio frequency (RF) signal, and performing other operations for extracting raw data from the received RF signal.

(87) FIG. 2 illustrates an example computer-implemented method for the claimed broadcast encryption scheme. In the illustrated example embodiment, setup routine generates master public key (MPK) **270** and provides it to the broadcasting authority **210**. The key operations **245**, **255**, **265** may be performed at a centralized or trusted authority or third-party, which may or may not be associated with or controlled by broadcast authority **210**. The setup routine **245** also generates the master secret key (MSK) **250**. Setup routine **245** may be called by the private key generator (PKG) **255**. PKG **255** outputs system master public-key MPK **250** and the system master secret-key MSK **250**, and makes MPK publicly available and keeps MSK as a secret. Key generation routine **265** receives the MPK and MSK, and user identities **275**, and outputs secret keys **280** for each

specific user.

(88) Broadcasting authority **210** then employs MPK **270** to perform an encryption **215**, which is then used as the ciphertext for a broadcast message **220**. The broadcast message is then provided over a broadcast channel **225**, which as described herein can take any wired or wireless form. The ciphertext is received by subscribed receivers **230** who have been provided with certain key material **280** in association with their identities **275** which have been provided to a key generation module **265**. The secret keys **280** can be provided to receivers based on their identities **275**. With the secret keys, the subscribed receivers can perform a decryption **235** of the broadcast ciphertext, and generate a resulting broadcast message **240**.

(89) FIGS. **3** and **4** depict example computer systems useful for implementing various embodiments described in the present disclosure. Various embodiments may be implemented, for example, using one or more computer systems, such as computer system **500** shown in FIG. **3**. One or more computer system(s) **500** may be used, for example, to implement any of the embodiments discussed herein, as well as combinations and sub-combinations thereof.

(90) Computer system **500** may include one or more processors (also called central processing units, processing devices, or CPUs), such as a processor **504**. Processor **504** may be connected to a communication infrastructure **506** (e.g., such as a bus).

(91) Computer system **500** may also include user input/output device(s) **503**, such as monitors, keyboards, pointing devices, etc., which may communicate with communication infrastructure **506** through user input/output interface(s) **502**. One or more of processors **504** may be a graphics processing unit (GPU). In an embodiment, a GPU may be a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.

(92) Computer system **500** may also include a main memory **508**, such as random-access memory (RAM). Main memory **508** may include one or more levels of cache. Main memory **508** may have stored therein control logic (i.e., computer software, instructions, etc.) and/or data. Computer system **500** may also include one or more secondary storage devices or secondary memory **510**. Secondary memory **510** may include, for example, a hard disk drive **512** and/or a removable storage device or removable storage drive **514**. Removable storage drive **514** may interact with a removable storage unit **518**. Removable storage unit **518** may include a computer-usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage drive **514** may read from and/or write to removable storage unit **518**.

(93) Secondary memory **510** may include other means, devices, components, instrumentalities, or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **500**. Such means, devices, components, instrumentalities, or other approaches may include, for example, a removable storage unit **522** and an interface **520**. Examples of the removable storage unit **522** and the interface **520** may include a program cartridge and cartridge interface, a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.

(94) Computer system **500** may further include communications interface **524** (e.g., network interface). Communications interface **524** may enable computer system **500** to communicate and interact with any combination of external devices, external networks, external entities, etc. (individually and collectively referenced as remote device(s), network(s), entity(ies) **528**). For example, communications interface **524** may allow computer system **500** to communicate with external or remote device(s), network(s), entity(ies) **528** over communications path **526**, which may be wired and/or wireless (or a combination thereof), and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **500** via communications path **526**.

(95) Computer system **500** may also be any of a personal digital assistant (PDA), desktop workstation, laptop or notebook computer, netbook, tablet, smartphone, smartwatch or other wearable devices, appliance, part of the Internet-of-Things, and/or embedded system, to name a few non-limiting examples, or any combination thereof.

(96) Computer system **500** may be a client or server computing device, accessing or hosting any applications and/or data through any delivery paradigm, including but not limited to remote or distributed cloud computing solutions; local or on-premises software (“on-premise” cloud-based solutions); “as a service” models (e.g., content as a service (CaaS), digital content as a service (DCaaS), software as a service (SaaS), managed software as a service (MSaaS), platform as a service (PaaS), desktop as a service (DaaS), framework as a service (FaaS), backend as a service (BaaS), mobile backend as a service (MBaaS), infrastructure as a service (IaaS), etc.); and/or a hybrid model including any combination of the foregoing examples or other services or delivery paradigms.

(97) FIG. **4** illustrates an example machine of a computer system **900** within which a set of instructions, for causing the machine to perform any one or more of the operations discussed herein, may be executed. In alternative implementations, the machine may be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, and/or the Internet. The machine may operate in the capacity of a server or a client machine in a client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment.

(98) The machine may be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, a specialized application or network security appliance or device, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

(99) The example computer system **900** includes a processing device **902**, a main memory **904** (e.g., read-only memory (ROM), flash memory, dynamic random-access memory (DRAM) such as synchronous DRAM (SDRAM), etc.), a static memory **906** (e.g., flash memory, static random-access memory (SRAM), etc.), and a data storage device **918**, which communicate with each other via a bus **930**.

(100) Processing device **902** represents one or more processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device may be complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device **902** may also be one or more special-purpose processing devices such as an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device **902** is configured to execute instructions **926** for performing the operations and steps discussed herein.

(101) The computer system **900** may further include a network interface device **908** to communicate over the network **920**. The computer system **900** also may include a video display unit **910**, an alphanumeric input device **912** (e.g., a keyboard), a cursor control device **914** (e.g., a mouse), a graphics processing unit **922**, a signal generation device **916** (e.g., a speaker), graphics processing unit **922**, video processing unit **928**, and audio processing unit **932**.

(102) The data storage device **918** may include a machine-readable medium **924** (also known as a computer-readable storage medium) on which is stored one or more sets of instructions **926** (e.g., software instructions) embodying any one or more of the operations described herein. The instructions **926** may also reside, completely or at least partially, within the main memory **904** and/or within the processing device **902** during execution thereof by the computer system **900**, where the main memory **904** and the processing device **902** also constitute machine-readable storage media.

(103) In an example, the instructions **926** include instructions to implement operations and functionality corresponding to the disclosed subject matter. While the machine-readable storage medium **924** is shown in an example implementation to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches

term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions 926 for execution by the machine and that cause the machine to perform any one or more of the operations of the present disclosure. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

(104) Some portions of the detailed description have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

(105) It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, discussions utilizing terms such as “identifying” or “determining” or “executing” or “performing” or “collecting” or “creating” or “sending” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage devices.

(106) The present disclosure also relates to an apparatus for performing the operations herein. This apparatus may be specially constructed for the intended purposes, or it may comprise a computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a computer-readable storage medium, such as but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

(107) The operations and illustrations presented herein are not inherently related to any particular computer or other apparatus. Various types of systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the operations. The structure for a variety of these systems will appear as set forth in the description herein. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the disclosure as described herein.

(108) The present disclosure may be provided as a computer program product, or software, that may include a machine-readable medium having stored thereon instructions, which may be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as read-only memory (“ROM”), random access memory (“RAM”), magnetic disk storage media, optical storage media, flash memory devices, etc.

(109) In some embodiments, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon may also be referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system 500, main memory 508, secondary memory 510, and removable storage units 518 and 522, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system 500), may cause such data processing devices to operate as described herein.

(110) Based on the teachings contained in this disclosure, it will be apparent to persons skilled in the relevant art(s) how to make and use embodiments of this disclosure using data processing devices, computer systems, and/or computer architectures other than that shown in FIGS. 3 and 4. In particular, embodiments can operate with software, hardware, and/or operating system implementations other than those described herein.

(111) It is to be appreciated that the Detailed Description section, and not any other section, is intended to be used to interpret the claims. Other sections can set forth one or more but not all exemplary embodiments as contemplated by the inventor(s), and thus, are not intended to limit this disclosure or the appended claims in any way.

(112) While this disclosure describes exemplary embodiments for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other embodiments and modifications thereto are possible and are within the scope and spirit of this disclosure. For example, and without limiting the generality of this paragraph, embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.

(113) Embodiments have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative embodiments can perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein.

(114) References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other embodiments whether or not explicitly mentioned or described herein. Additionally, some embodiments can be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, some embodiments can be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

(115) The breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments but should be defined only in accordance with the following claims and their equivalents. In the foregoing specification, implementations of the disclosure have been described with reference to specific example implementations thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of implementations of the disclosure as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A method for encrypting a message for transmitting over a broadcast network to multiple recipients as a ciphertext, the method comprising: receiving an authorized recipient set S at a broadcast module, wherein the authorized recipient set identifies a set of recipients with permission to recover a broadcast message, wherein S is a subset of {1, 2, . . . , N}, wherein n is an integer such that $n \geq N$, and wherein N is the total number of possible recipients of the message; generating a public key mpk at an authority, the public key mpk comprising: a) a pairing group, the pairing group further comprising a first source group, a second source group, and a target group, b) vectors $[v]_{\text{sub.1}}$, $[w]_{\text{sub.1}}$ and $[w]_{\text{sub.2}}$ as n random group elements from the first source group, c) a random group element $[v]_{\text{sub.0}}$ from the first source group, and d) a random group

element a of $[a].sub.T$ from the target group; receiving the public key mpk at a broadcast module from the authority; receiving or creating a message M for encryption; encoding the authorized recipient set S as a degree 3 polynomial, wherein the degree 3 polynomial has a coefficient vector f equal to a characteristic vector of the recipient set S , and wherein the vector f has a length $n.sub.3$; cryptographically compressing the vector f using the vector $[w.sub.2].sub.1$ of n group elements from the public key mpk and by sampling a random vector $w.sub.3$ of n integers modulo p resulting in a digest of n group elements; generating a ciphertext of message M by: (a) sampling a random scalar s ; (b) multiplying the random scalar s with the digest, the vector $w.sub.3$ and the public key mpk ; (c) concatenating the output of the multiplying to form the ciphertext; (d) multiplying the random scalar s by $[a].sub.T$ to derive $[as].sub.T$; (e) masking message M with $[as].sub.T$; and transmitting the masked message as a ciphertext and the authorized recipient set S to recipients on the broadcast network.

2. The method of claim 1, wherein generating the ciphertext is performed by executing: parsing $mpk = ([w.sub.1].sub.1, [w.sub.2].sub.1, [v.sub.0].sub.1, [v].sub.1, [a].sub.T)$; sampling $s \leftarrow \text{custom character.sub.p, } w.sub.3 \leftarrow \text{custom character.sub.p.sup.n}$; and outputting ciphertext $ct = [s].sub.1, [(I.sub.n.Math.w.sub.2.Math.w.sub.3)f.sup.T + w.sub.1.sup.T]s.sub.1, [w.sub.3s].sub.1, [(v.sub.0w.sub.3 + v)s].sub.1, [as].sub.T.Math.M$.

3. The method of claim 1, further comprising executing instructions for decrypting the ciphertext by a recipient to recover a broadcast message, the instructions configured for: receiving the ciphertext and the authorized recipient set S at a receiver module from the remote broadcast module over the broadcast network, wherein the ciphertext further comprises a digest, and the digest further comprises information about the polynomial; receiving a secret key from a remote authority module, the secret key comprising group elements from the second source group; checking whether the recipient is in a set of authorized recipients; computing the degree 3 polynomial that has a coefficient vector f equal to a characteristic vector of the authorized recipient set S , wherein the degree 3 polynomial has a coefficient vector f equal to a characteristic vector of the authorized recipient set S , and wherein the vector f has a length $n.sub.3$; recovering the broadcast message by evaluating the polynomial with the digest using the group elements in a pairing algorithm; and storing the recovered broadcast message for further processing at the receiver module.

4. A system for encrypting a message for transmitting over a broadcast network to multiple recipients as a ciphertext, comprising: a network communication interface configured to communicate over a wired or wireless network; a set of processing circuits coupled to the network communication interface, the processing circuits comprising modules configured for: receiving an authorized recipient set S , wherein the authorized recipient set identifies a set of recipients with permission to recover a broadcast message, wherein S is a subset of $\{1, 2, \dots, N\}$, wherein n is an integer such that $n.sub.3 \geq N$, and wherein N is the total number of possible recipients of the message; generating a public key mpk at an authority, the public key mpk comprising: a) a pairing group, the pairing group further comprising a first source group, a second source group, and a target group, b) vectors $[v].sub.1, [w.sub.1].sub.1$ and $[w.sub.2].sub.1$ as n random group elements from the first source group, c) a random group element $[v.sub.0].sub.1$ from the first source group, and d) a random group element $[a].sub.T$ from the target group; receiving the public key mpk from the authority; receiving or creating a message M for encryption; encoding the authorized recipient set S as a degree 3 polynomial, wherein the degree 3 polynomial has a coefficient vector f equal to a characteristic vector of the recipient set S , and wherein the vector f has a length $n.sub.3$; cryptographically compressing the vector f using the vector $[w.sub.2].sub.1$ of n group elements from the public key mpk and by sampling a random vector $w.sub.3$ of n integers modulo p resulting in a digest of n group elements; generating a ciphertext of message M by: (a) sampling a random scalar s ; (b) multiplying the random scalar s with the digest, the vector $w.sub.3$ and the public key mpk ; (c) concatenating the output of the multiplying to form the ciphertext (d) multiplying the random scalar s by $[a].sub.T$ to derive $[as].sub.T$; (e) masking message M with $[as].sub.T$; and transmitting the masked message as a ciphertext and the authorized recipient set S to recipients on the broadcast network S .

5. The system of claim 4, wherein generating the ciphertext is performed by executing: parsing $mpk = ([w.sub.1].sub.1, [w.sub.2].sub.1, [v.sub.0].sub.1, [v].sub.1, [a].sub.T)$; sampling $s \leftarrow \text{custom character.sub.p, } w.sub.3 \leftarrow \text{custom character.sub.p.sup.n}$; and outputting ciphertext $ct = [s].sub.1, [(I.sub.n.Math.w.sub.2.Math.w.sub.3)f.sup.T + w.sub.1.sup.T]s.sub.1, [w.sub.3s].sub.1, [(v.sub.0w.sub.3 + v)s].sub.1, [as].sub.T.Math.M$.

6. The system of claim 4, further configured for decrypting a ciphertext by a recipient to recover a broadcast message, including: receiving the ciphertext and the authorized recipient set S at a receiver module from a remote broadcast module over the broadcast network, wherein the ciphertext further comprises a digest, and the digest further comprises information about the polynomial; receiving a secret key, the secret key comprising group elements from the second source group; checking whether the recipient is in a set of authorized recipients; computing the degree 3 polynomial that has a coefficient vector f equal to a characteristic vector of the authorized recipient set S ; wherein the degree 3 polynomial has a coefficient vector f equal to a characteristic vector of the authorized recipient set S , and wherein the vector f has a length $n.sub.3$; recovering the broadcast message by evaluating the polynomial with the digest using the group elements in a pairing algorithm; and storing the recovered broadcast message for further processing at the receiver module.

7. A non-transitory computer-readable storage medium storing computer-executable instructions that, when executed by one or more processors of a computing device, configure the one or more processors to perform instructions for encrypting a message for transmitting over a broadcast network to multiple recipients as a ciphertext, the instructions comprising: receiving an authorized recipient set S at a broadcast module, wherein the authorized recipient set identifies a set of recipients with permission to recover a broadcast message, wherein S is a subset of $\{1, 2, \dots, N\}$, wherein n is an integer such that $n.sub.3 \geq N$, and wherein N is the total number of possible recipients of the message; generating a public key mpk at an authority, the public key mpk comprising: a) a pairing group, the pairing group further comprising a first source group, a second source group, and a target group, b) vectors $[v].sub.1, [w.sub.1].sub.1$ and $[w.sub.2].sub.1$ as n random group elements from the first source group, c) a random group element $[v.sub.0].sub.1$ from the first source group, and d) a random group element $[a].sub.T$ from the target group; receiving the public key mpk at the broadcast module from the authority; receiving or creating a message M for encryption; encoding the authorized recipient set S as a degree 3 polynomial, wherein the degree 3 polynomial has a coefficient vector f equal to a characteristic vector of the recipient set S , and wherein the vector f has a length $n.sub.3$; cryptographically compressing the vector f using the vector $[w.sub.2].sub.1$ of n group elements from the public key mpk and by sampling a random vector $w.sub.3$ of n integers modulo p resulting in a digest of n group elements; generating a ciphertext of message M by: (a) sampling a random scalar s ; (b) multiplying the random scalar s with the digest, the vector $w.sub.3$ and the public key mpk ; (c) concatenating the output of the multiplying to form the ciphertext; (d) multiplying the random scalar s by $[a].sub.T$ to derive $[as].sub.T$; (e) masking message M with $[as].sub.T$; and transmitting the masked message as a ciphertext and the authorized recipient set S to recipients on the broadcast network.

8. The non-transitory computer-readable storage medium of claim 7, wherein generating the ciphertext is performed by executing: parsing $mpk = ([w.sub.1].sub.1, [w.sub.2].sub.1, [v.sub.0].sub.1, [v].sub.1, [a].sub.T)$; sampling $s \leftarrow \text{custom character.sub.p; } w.sub.3 \leftarrow \text{custom character.sub.p.sup.n}$; and outputting ciphertext $ct = [s].sub.1, [(I.sub.n.Math.w.sub.2.Math.w.sub.3)f.sup.T + w.sub.1.sup.T]s.sub.1, [w.sub.3s].sub.1, [(v.sub.0w.sub.3 + v)s].sub.1, [as].sub.T.Math.M$.

9. The non-transitory computer-readable storage medium of claim 7, further comprising executing instructions for decrypting the ciphertext by a recipient to recover a broadcast message, the instructions configured for: receiving the ciphertext and the authorized recipient set S at a receiver module from a remote broadcast module over the broadcast network, wherein the ciphertext further comprises a digest, and the digest further comprises information about the polynomial; receiving a secret key from a remote authority module, the secret key comprising group elements from the second source group; checking whether the recipient is in a set of authorized recipients; computing the degree 3 polynomial that has a coefficient vector f equal to a characteristic vector of the authorized recipient set S ; wherein the degree 3 polynomial has a coefficient vector f equal to a characteristic vector of the authorized recipient set S , and wherein the vector f has a length $n.sub.3$; recovering the broadcast message by evaluating the polynomial with the digest using the group elements in a pairing algorithm; and storing the recovered broadcast message for further processing at the receiver module.